

Pratyush Padhy

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EDUCATION

University of California, Irvine

Irvine, CA

B.S. Computer Science, Specialization in Intelligent Systems, GPA: 3.8/4.0

Expected: 2028

- Dean's Honor List: Fall 2025, Winter 2026, Spring 2026 | Orgs: Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

EXPERIENCE

Amazon Web Services (AWS)

East Palo Alto, CA

Incoming Software Development Engineer Intern

Sept. 2026 – Dec. 2026

- Joining the Open Data Analytics Engines Runtime team, which owns the Spark, Trino, and Flink query engines

Tata Consultancy Services (TCS)

Pittsburgh, PA

AI/ML Intern, Medical Robotics Center

Jun. 2026 – Aug. 2026

- Benchmarked reinforcement learning against a pretrained vision model over **250+** liquid-pouring trials
- Reached **85.3% task success** using camera input alone, ahead of the **81.5% baseline** that had full sensor data
- Trained that baseline to **81.5% success** across 3 seeds by reparameterizing how the robot's actions are encoded
- Isolated pretrained vision as the deciding cause by retraining from scratch on identical input, which scored **0%**
- Raised pick-and-place success to **10/10** at **3.7cm** error, up from **0/10** at **63cm**, by tuning joint control and IK

Ready Tutor

Irvine, CA

Software Developer Intern

Mar. 2026 – Jun. 2026

- Built the real-time course search across **1,200+ university courses**, the headline student-facing feature
- Anchored the funding demo with that search, directly helping the team secure **\$5,500** for the product
- Cut database writes to **1 per user action** by consolidating **10+** scattered storage calls into one shared layer
- Shipped **30+ production fixes**, including a credit-calculation bug that was breaking student schedule planning
- Replaced a rigid **4-year** course planner with an unlimited-year layout built on reusable per-quarter blocks

PROJECTS

Chronicle | Python, FastAPI, PostgreSQL, Next.js, Docker | chronicles-weld.vercel.app

Jun. 2026 – Sept. 2026

- Job search tool indexing **36,000+ open roles** from **600+** verified company ATS boards, refreshed daily
- Returned ranked search results in **19ms**; personalized ranking lifted result quality **40%** over keyword search
- Eliminated duplicate postings by fixing a bug that counted a single role once for every city it was listed in
- Cut stored data by **126 MB** by stripping raw ATS HTML, keeping the corpus inside a **512 MB** limit
- Shipped saved searches, a kanban application tracker, and email alerts, all gated by **119 automated tests**

Forge | Python, PyTorch, Docker, Ollama | forge-grpo.vercel.app

Jul. 2026 – Sept. 2026

- Training pipeline teaching a **1.5 billion parameter** open-source model to reason with no human-labeled data
- Raised reasoning accuracy from **58.8% to 70.0%** on 1,319 held-out math problems using reinforcement learning
- Compressed the trained model **3.15x** through quantization and served it live behind a base-vs-tuned playground
- Diagnosed a zero-reward cold start that stalled training, then verified under **1%** drop on a held-out set

FacePulse | Python, TensorFlow, FastAPI, React, OpenCV | face-pulse.netlify.app

Apr. 2026 – Jun. 2026

- Web app that classifies facial expressions into **5 emotion classes** from a live webcam feed in real time
- **Featured on the Handshake AI Showcase**, selected from among thousands of nationwide student entries
- Compared 4 model designs on a **35,000 image** dataset, with the custom network leading at **64.5% accuracy**
- Built the pipeline end to end, from raw labeled image data through training to fully deployed inference

TECHNICAL SKILLS

Languages: Python, Java, SQL (Postgres), JavaScript, TypeScript, HTML/CSS, Bash

Frameworks: PyTorch, TensorFlow, FastAPI, React, Next.js, scikit-learn, Stable-Baselines3, Gymnasium, SQLAlchemy

Developer Tools: Git, GitHub Actions, Docker, pytest, ROS2, Gazebo, WSL2/CUDA, AWS, Google Cloud Run, Vercel

Libraries: NumPy, Pandas, Matplotlib, OpenCV, Transformers, Ollama, vLLM

Databases: PostgreSQL, pgvector, MongoDB, SQLite, Neon